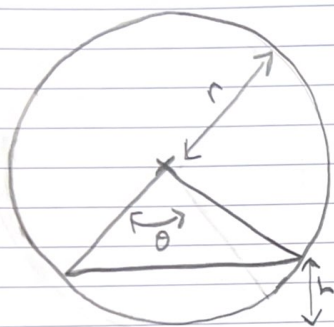
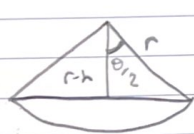


Equation 1 - Derivation



Area of a sector:

$$A = \frac{1}{2} \theta r^2$$

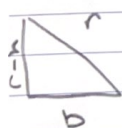


$$\Rightarrow \frac{\theta}{2} = \cos^{-1}\left(\frac{r-h}{r}\right)$$

$$\Rightarrow \theta = 2 \cos^{-1}\left(\frac{r-h}{r}\right)$$

Area of sector:

$$A = \frac{1}{2} \times 2 \cos^{-1}\left(\frac{r-h}{r}\right) r^2 \Rightarrow \cos^{-1}\left(\frac{r-h}{r}\right)$$



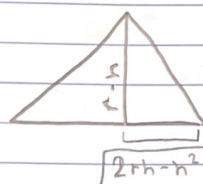
$$\Rightarrow r^2 = (r-h)^2 + b^2$$

$$b^2 = r^2 - (r-h)^2$$

$$b^2 = r^2 - (r^2 - 2rh + h^2)$$

$$b^2 = 2rh - h^2$$

$$\Rightarrow b = \sqrt{2rh - h^2}$$



Area of triangle:

A = height x base

$$(r-h) \sqrt{2rh - h^2} = \text{Area}$$

Area of a segment:

Area of sector - Area of triangle = A

$$\Rightarrow \text{Area} = \cos^{-1}\left(\frac{r-h}{r}\right) r^2 - (r-h) \sqrt{2rh - h^2}$$

For volume of cylinder at height h, need to time by length:

$$V(h) = \left[\cos^{-1}\left(\frac{r-h}{r}\right) r^2 - (r-h) \sqrt{2rh - h^2} \right] \times L$$

where r = radius, h = height of segment and L = length of cylinder